Kuruva Jeshwanth Krishna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.34
Sri Chaitanya school	Class XII (STATE)	2024	97.20%
Sri Chaitanya school	Class X (STATE)	2022	9.20%

SCHOLASTIC ACHIEVEMENTS

- Finished **4th in Optiver Trade-A-Thon** held in Feb 2025
- Rank **122** ICPC preliminary round 2025
- NSEP:Among **top 1%** performers in Telangana in NSEP 2023
- Amongst the center **top 10%** performers in NSEA,NSEC and NSEP 2023

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant, Probability and Random Variables (AI1110):** Assisted in grading assignments and exams, and conducted multiple in-class tutorial sessions for a cohort of 160 students.
- **Core, Infero:** Assisted in creating and validating competitive programming problems and supported teaching sessions on competitive programming concepts.

SKILLS

- **Programming Languages**: C,C++,python,Java,Shell script,Assembly(RISC V)
- **Tools:**Git/Github,Jupyter notebooks,SolidEdge

RELEVANT COURSES

- **Mathematics**: Discrete Maths, Linear Algebra, Probability and Random variables
- **Computer science:** Data structures, Algorithms ,Computer architecture, Software Development Fundamentals,Database and Management systems, Operating systems, Theory of Computation
- **Electrical and Electronics**: Digital Circuits ,Basic Electrical Engineering

PROJECTS

xv6 Kernel Extensions | *C, xv6 OS, X86 Architecture github*

Operating Systems Project | January 2026 - April 2026

- **CPU Scheduling & Core OS**: Enhanced the xv6 kernel with custom system calls, process accounting utilities, and a Multi-Level Feedback Queue (MLFQ) scheduler supporting priority-based preemption, time-slice management, and starvation prevention.
- **Virtual Memory Architecture**: Architected a disk-backed virtual memory subsystem featuring lazy allocation, page-fault driven memory recovery, page eviction, and swap-in/swap-out handling to enable program execution beyond physical memory limits.
- **Multi-Disk Storage & RAID**: Integrated configurable disk scheduling algorithms for swap I/O alongside RAID 0,

RAID 1, and RAID 5 multi-disk storage policies to optimize swap block allocation and fault-tolerant recovery.
- **Instrumentation & Evaluation**: Implemented kernel-level telemetry to collect real-time memory, swap, and disk metrics, leveraging workload-based performance measurements to analyze I/O bottlenecks and paging activity.

FPU-Assisted Optimization of Integer Workloads using gem5 *(Computer Architecture Project)*

Tech Stack: C, gem5, RISC-V, Computer Architecture, Performance Analysis

- Investigated the use of **Floating Point Units (FPUs)** to accelerate integer-intensive workloads on an **out-of-order RISC-V processor** simulated using **gem5**.
- Designed and benchmarked optimized implementations of **matrix multiplication**, **dot product**, **determinant computation (Bareiss algorithm)**, **minimum element search**, and **factorial modulo**, comparing **integer-only** and **mixed integer/FPU execution**.
- Applied **loop unrolling**, **instruction-level parallelism (ILP)**, **cache-aware loop ordering (IKJ)**, **functional-unit scheduling**, and **mixed integer/FPU execution** based on pipeline latencies and functional unit availability.
- Analyzed **cache behavior**, **branch prediction effects**, and **instruction scheduling trade-offs** using execution statistics collected from **gem5** simulations.
- Achieved workload-dependent performance improvements of **15-30% median speedup**, with the best-performing optimizations exceeding **30%** over optimized integer baselines.

Infinite Precision Calculator | *Java, Apache Ant, Python, Git github*

Software Development Fundamentals Course | April 2025 - May 2025

- **Developed aarithmetic.jar**: Built a custom Java library enabling arbitrary-precision integer and 30-decimal float operations by implementing string-based storage to bypass primitive data type limits.
- **Algorithmic Design & Automation**: Engineered core digit-by-digit arithmetic logic and streamlined edge-case validation using automated Python verification scripts and Apache Ant build configurations.

EXTRACURRICULAR

- **Competitive Programming:** Peak **Codeforces Specialist (1543)** and **CodeChef 3star (1654)**; solved **900+ algorithmic problems** across Codeforces, CodeChef, other competitive programming platforms.
- **NSO Cricket (2024-2028):** Represented the institute in cricket as part of the **National Sports Organization**
- **Core Member, Otaku - Anime Club, IITH:** Coordinated and executed **cosplay events, anime quizzes, and screening sessions**, contributing to event planning, logistics, and community engagement.